

Java Practice Work – Encapsulation (Medium Level)

Question: Product Details using Encapsulation

Write a Java program to create a Product class using encapsulation. All variables must be private and accessed only through getters and setters.

No	Requirement
1	Create class Product
2	Private variables: String productName, int productId, double price, int quantity
3	Create setter methods for all variables
4	Create getter methods for all variables
5	Validate in setters: name not empty, id > 0, price > 0, quantity >= 0
6	addStock(int amount) → increase quantity if amount > 0
7	sellProduct(int amount) → decrease quantity if amount > 0 and <= quantity
8	applyDiscount(double percent) → apply if percent > 0 and < 100
9	calculateTotalValue() → return price * quantity
10	displayProductDetails() → print all details
11	In main(): create object, set values, call all methods, print final result

Sample Input

Product Name: Laptop Product Id: 101 Price: 50000 Quantity: 10 Add Stock: 5 Sell Product: 3 Discount: 10%

Expected Output

Product Details Product Name: Laptop Product Id: 101 Price: 45000.0 Quantity: 12 Total Stock Value: 540000.0

Explanation

Initial quantity = 10 → after adding 5 = 15 → after selling 3 = 12. Initial price = 50000 → after 10% discount = 45000. Total value = 45000 × 12 = 540000.0